

4th Homework (due Oct 14)

k always denotes an algebraically closed field of characteristic 0 and varieties are over k .

Problem 1 (Existence of smooth points, 3P + 3P + 4P + 3P)

Prove that every variety has a smooth point.

(In the following, you may take inspiration from our proof of images being constructible.)

- (a) First reduce to the affine irreducible case, i.e. to an integral domain finite type k -algebra A .
- (b) Using Noether normalization and the fact that compositions of smooth maps are smooth, reduce to showing that for a finite extension of integral domains $R \subseteq A$, the map $\mathrm{Sp}(A) \rightarrow \mathrm{Sp}(R)$ has a smooth point.
- (c) To show $\mathrm{Sp}(A) \rightarrow \mathrm{Sp}(R)$ is smooth in some point, first consider the case $A = R[\alpha]$ is generated by a single element. Use that $\mathrm{Quot}(A)$ is a finite *separable* extension of $\mathrm{Quot}(R)$ because of the $\mathrm{char}(k) = 0$ assumption.
- (d) Wrap up the proof by extending to the general case.

Problem 2 (Orbits in $M_n(k)$, 3P + 5P)

- (a) Let G be a LAG acting on a variety X . Let $x, y \in X$ be two points and let $Z \subseteq X$ be a closed subset such that

$$Gy \cap \overline{(Z \cap Gx)} \neq \emptyset.$$

Show that then already $Gy \subseteq \overline{Gx}$.

- (b) Given $n \geq 1$ and $\lambda \in k$, consider the variety

$$X_\lambda = \{x \in M_n(k) \mid \mathrm{charpoly}(x, T) = (T - \lambda)^n\}.$$

Let $G = \mathrm{GL}_n$ act on X_λ by conjugation. Describe the G -orbits of X_λ combinatorially, determine their dimensions, and describe all their closure relations.